PHYS 143 – Mechanics and Thermal Physics

Temperature and Heat

What is temperature, how do we measure it, and how does thermal energy move?

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By the end of this week, you should be able to:

1. Distinguish temperature from heat, and explain thermal equilibrium and the zeroth law.
2. Convert between the Celsius, Fahrenheit and Kelvin scales and describe how thermometers work.
3. Calculate thermal expansion and the heat needed to change a temperature, including simple calorimetry.
4. Calculate the rate of heat transfer by conduction and radiation, and describe convection.

Central question

What is temperature, how do we measure it, and how does thermal energy move?

The thread of the week

temperature → **equilibrium** → **scales** → **expansion** → $Q = mc\Delta T$ → **calorimetry** → **conduction, convection, radiation.** Convert units first, predict the sign, then calculate.

Two contacts: A (Tuesday) **DEFINE** temperature and heat, then **MODEL** what a temperature change does (expansion and $Q = mc\Delta T$); **B** (Wednesday) **MODEL** calorimetry, then **APPLY** conduction, convection and radiation to combined problems.

Reading: Young and Freedman, Chapter 17, §17.1–17.4 and §17.6. **Labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

Not marked. Answer individually, then check the answers.

1. Convert $2.5\,\mathrm{kW}$ to watts and $3.0\,\mathrm{cm}^2$ to m^2.
2. How much energy does a $100\,\mathrm{W}$ lamp use in 2.0 minutes?
3. A $2.0\,\mathrm{m}$ steel bar grows by 0.0010%. Find its length increase.
4. Solve $5(T-32)/9 = 37$ for T.
5. Find the volume of a cube of side $0.10\,\mathrm{m}$, in m^3.
6. Two bodies exchange energy. If one gains $300\,\mathrm{J}$, how much does the other lose?

$2500\,\mathrm{W}$; $3.0\,\mathrm{cm}^2 = 3.0 \times 10^{-4}\,\mathrm{m}^2$.

$E = Pt = (100)(120) = 1.2 \times 10^4\,\mathrm{J}$.

$0.0010\% = 1.0 \times 10^{-5}$, so $\Delta L = (1.0 \times 10^{-5})(2.0) = 2.0 \times 10^{-5}\,\mathrm{m}$.

$T - 32 = 66.6 \Rightarrow T = 98.6$.

$(0.10)^3 = 1.0 \times 10^{-3}\,\mathrm{m}^3$.

It loses $300\,\mathrm{J}$ (energy is conserved).

Units and powers of ten (Lecture 1), energy and power (Lectures 4 and 5) are reused. Thermal physics adds one new quantity, temperature, and one new way energy is exchanged, heat.

Contact A: Define, then Model

What are temperature and heat, how do we measure temperature, and what does a temperature change do?

Poll

A metal spoon and a wooden spoon have been in the same kitchen all day. You touch both. The metal one feels colder. **Which spoon has the lower temperature?**

Predict first

(a) the metal one, (b) the wooden one, (c) both are at the same temperature. What does your hand actually sense: the temperature of the spoon, or how fast energy leaves your hand?

(c). After a day in the same room both are in equilibrium with the air. Metal *conducts* energy away from your hand faster, so it *feels* colder. We return to this in Contact B.

Defining the vocabulary of temperature and heat

CORE

Quantity	What it is	What it measures	SI unit	Representation
Temperature T	the property of a body's state that decides which way energy flows	how hot or cold, on a chosen scale	K (also $^\circ\mathrm{C}$, $^\circ\mathrm{F}$)	a thermometer reading; a point on a scale
Heat Q	energy transferred between bodies because of a temperature difference	how much energy has moved	J	an arrow from the hotter to the cooler body
Thermal energy	the energy of the random motion of the particles	the energy stored inside a body	J	not drawn as a flow

Interpret: temperature answers *which way will energy flow?* Heat answers *how much energy moved?*

A body has temperature and thermal energy, but not "heat".

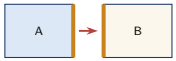
- A **thermodynamic system** is the part of the universe we study; the rest is the **surroundings**.
- Its **state** is described by macroscopic variables: volume, pressure, mass, temperature.
- **Intensive** variables do not depend on the amount (temperature, pressure); **extensive** ones do (mass, volume).
- A **process** takes the system from one state to another.

Physical meaning

We describe the system with a few measurable, large-scale quantities, without tracking individual molecules. Temperature is one of these.

Predict first

Mass and temperature of a water sample: which is intensive? What happens to each if you split the sample in two?



diathermal wall:
energy can pass



adiabatic wall:
no energy passes

Rule: thermal equilibrium

Two systems separated by a **diathermal** wall exchange energy until there is no *net* exchange: they are in **thermal equilibrium** and have the same temperature. An **adiabatic** wall prevents exchange.

Introduce. Place two bodies at different temperatures in contact for a long time. Energy (heat) passes from the hotter to the cooler body until they reach a common temperature.

Explain. A diathermal wall (thin metal, for example) lets energy cross; an adiabatic wall (thick foam, for example) does not. At equilibrium energy still crosses the wall in both directions, but the two flows are equal.

Predict: if the wall between the bodies is adiabatic, do their temperatures ever become equal?

Temperature T

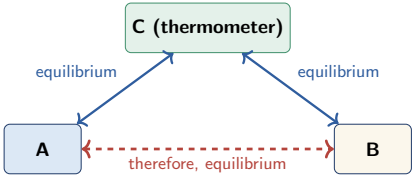
A property of the *state* of a body. It tells which way energy will flow when two bodies touch: from the higher to the lower temperature.

Heat Q

Energy *transferred* between bodies because of a temperature difference. Unit: joule.

Common errors

(1) "The body contains heat." A body has thermal (internal) energy; heat is the energy *in transit*.
(2) "Cold flows in." Energy flows out of the warmer body.
(3) "Equilibrium means no exchange." It means no *net* exchange.



Introduce. Take three bodies A, B and C, initially at different temperatures. Let C touch A and then B, until each is in equilibrium with C.

Rule: the zeroth law

If A and B are each in thermal equilibrium with C, then A and B are in thermal equilibrium with each other.

Physical meaning

This is why thermometers work: C is the thermometer. If it reads the same when touching A and B, then A and B have the same temperature, even if they never touch.

- Choose a **thermometric property** that changes reproducibly with temperature.
- **Calibrate** it at fixed points (for example, melting ice and boiling water).
- Read the value when the thermometer is in equilibrium with the body.

Examples of thermometric properties:

- liquid in glass: length of the column;
- resistance thermometer: electrical resistance;
- constant-volume gas: gas pressure;
- infrared (temporal artery): radiated power.

Common error

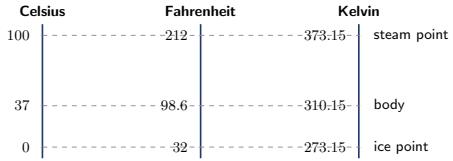
A thermometer reads *its own* temperature. It is only meaningful once it has reached equilibrium with the body.

Decide first

1. Can energy flow between two bodies at the same temperature?
2. You place a thermometer in hot tea and it reads $80^\circ\mathrm{C}$ immediately. Is that the tea's temperature?
3. Does a body at a higher temperature always contain more thermal energy?
4. Can a temperature be lower than $0\,\mathrm{K}$?

Answers

1. **No net flow.** Equal temperatures mean thermal equilibrium.
2. **Not until it reaches equilibrium.** A thermometer reads its own temperature.
3. **No.** A bathtub of warm water holds more thermal energy than a hot cup: energy also depends on the amount.
4. **No.** $0\,\mathrm{K}$ is the lower limit of the scale.



	K	°C	°F
water boils	373	100	212
water freezes	273	0	32
CO_2 solidifies	195	-78	-109
oxygen liquefies	90	-183	-298
absolute zero	0	-273	-460

Rule: scale conversions

$$T_F = \tfrac{9}{5}T_C + 32 \qquad T_C = \tfrac{5}{9}(T_F - 32)$$
$$T_K = T_C + 273.15$$

Define. Celsius reads 0 to $100^\circ\mathrm{C}$ between the freezing and boiling points of water; Fahrenheit reads 32 to $212^\circ\mathrm{F}$; the kelvin scale starts at $0\,\mathrm{K}$ and is written without a degree sign.

Explain. Absolute zero, $0\,\mathrm{K} = -273.15^\circ\mathrm{C}$, is the lower limit: a constant-volume gas thermometer extrapolates to zero pressure there.

- **TOOLKIT 1 – Temperature scales and conversions**

1. **IDENTIFY:** is the quantity a *temperature* T or a temperature *difference* ΔT?
2. **PREDICT** the size: a body at $37^\circ\mathrm{C}$ is about $98^\circ\mathrm{F}$ and $310\,\mathrm{K}$.
3. **CONVERT a temperature:** use the formulas above, in the right order.
4. **CONVERT a difference:** $\Delta T_K = \Delta T_C$ and $\Delta T_F = \frac{9}{5}\Delta T_C$; *no* offsets of 32 or 273.15.

RULE: use kelvin in any formula with T^4 or a ratio of temperatures; $^\circ\mathrm{C}$ is fine for ΔT.
WATCH OUT: adding 273.15 to a difference, or 32 to a Fahrenheit difference.
CHECK: is $T_K > 0$, and does the size agree with your prediction?

Question

You jump into a pool at $25.00^\circ\mathrm{C}$ and your body temperature is $37.00^\circ\mathrm{C}$. Express both temperatures in kelvin and in $^\circ\mathrm{F}$, and find the *change* $\Delta T = T_2 - T_1$ in $^\circ\mathrm{C}$, in K and in $^\circ\mathrm{F}$.

Predict first

Is the change in kelvin larger, smaller or equal to the change in $^\circ\mathrm{C}$? And in $^\circ\mathrm{F}$?

Temperature: $T_{1K} = 25.00 + 273.15 = 298.15\,\mathrm{K}$, $T_{2K} = 310.15\,\mathrm{K}$;
$T_{1F} = \frac{9}{5}(25.00) + 32 = 77.0^\circ\mathrm{F}$, $T_{2F} = 98.6^\circ\mathrm{F}$.
Change: $\Delta T = 12.00^\circ\mathrm{C} = 12.00\,\mathrm{K}$ (equal, no offset); $\Delta T_F = \frac{9}{5}(12.00) = 21.6^\circ\mathrm{F}$.
Evaluate: equal in K and $^\circ\mathrm{C}$, larger in $^\circ\mathrm{F}$, as predicted. A doctor who says "310 kelvin" is not alarming. ✓

Independent problem

(a) An oven is set to $180^\circ\mathrm{C}$. Express this in $^\circ\mathrm{F}$ and in K.
(b) Overnight the temperature falls from $12^\circ\mathrm{C}$ to $-3^\circ\mathrm{C}$. Find the change in K and in $^\circ\mathrm{F}$.

Predict first

Is $180^\circ\mathrm{C}$ larger or smaller than $180^\circ\mathrm{F}$? Is ΔT in (b) positive or negative?

(a) $T_F = \frac{9}{5}(180) + 32 = 356^\circ\mathrm{F}$; $\quad T_K = 180 + 273.15 = 453\,\mathrm{K}$.
(b) $\Delta T = -3 - 12 = -15^\circ\mathrm{C} = -15\,\mathrm{K}$; $\quad \Delta T_F = \frac{9}{5}(-15) = -27^\circ\mathrm{F}$.
Evaluate: $180^\circ\mathrm{C} > 180^\circ\mathrm{F}$ (since $180^\circ\mathrm{F} \approx 82^\circ\mathrm{C}$); the change is negative, as predicted. ✓

Common mistakes and the correct habit

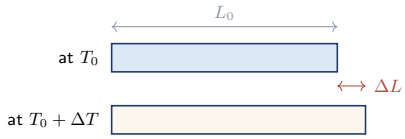
Mistake	Instead
Saying a body "contains heat" or that "cold flows in".	Heat is energy in transit; energy flows out of the hotter body.
Adding 273.15 or 32 to a temperature difference.	Differences have no offsets: $\Delta T_K = \Delta T_C$.
Reading a thermometer before equilibrium.	It reads its own temperature.
Saying equilibrium means no exchange.	It means no *net* exchange.
Equating a higher temperature with more thermal energy.	Thermal energy also depends on the amount.

Poll

Railway tracks and bridges have small gaps (expansion joints). What is the gap for, and how large must it be for a $12\,\mathrm{m}$ steel rail whose temperature varies by $40\,\mathrm{K}$?

Predict first

Order of magnitude: $0.1\,\mathrm{mm}$, $5\,\mathrm{mm}$ or $50\,\mathrm{mm}$?



Rule: linear expansion

$$\Delta L = \alpha L_0 \, \Delta T$$

α: coefficient of linear expansion (unit K^{-1}).

- Area: $\Delta A \approx 2\alpha A_0 \Delta T$. Volume: $\Delta V = \beta V_0 \Delta T$ with $\beta \approx 3\alpha$ for a solid.
- Valid for **small** ΔT and a uniform material; α depends slightly on temperature.

Coefficients of expansion

Linear expansion, α in K^{-1}

Material	α $(\times 10^{-5})$
aluminium	2.4
brass	2.0
copper	1.7
glass	0.4–0.9
invar (nickel–iron alloy)	0.09
quartz (fused)	0.04
steel	1.2

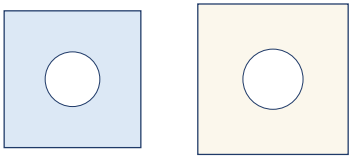
Volume expansion, β in K^{-1}

Solid	β $(\times 10^{-5})$	Liquid	β $(\times 10^{-5})$
aluminium	7.2	ethanol	75
brass	6.0	carbon disulfide	115
copper	5.1	glycerin	49
glass	1.2–2.7	mercury	18
invar	0.27		
quartz (fused)	0.12		
steel	3.6		

The unit K^{-1} equals $(^\circ\mathrm{C})^{-1}$. Check $\beta \approx 3\alpha$ on any solid, for example copper: $3(1.7) = 5.1$. Liquids expand far more than solids, which is how a liquid-in-glass thermometer works.

Does a hole expand or shrink?

CORE



cold plate

heated plate: hole grows too

Predict first

A metal plate with a circular hole is heated. Does the hole get smaller, larger, or stay the same?

Larger. Every length in the plate, including the diameter of the hole, scales by the same factor: the hole expands as if it were made of the same material.

But what if... you think "the metal grows into the hole"? Then the hole would shrink, and it does not.

- **TOOLKIT 2 – Thermal expansion**

1. IDENTIFY the quantity that changes (length, area, volume) and the material (α from a table).
2. ΔT is final minus initial; the sign gives expansion ($+$) or contraction ($-$).
3. CALCULATE: $\Delta L = \alpha L_0 \Delta T$ (or $\Delta V = \beta V_0 \Delta T$, $\beta \approx 3\alpha$).
4. COMPARE with the size of the object (is $\Delta L/L_0$ small?).
5. THERMAL STRESS: if expansion is prevented, the stress is $F/A = Y\alpha\Delta T$ ($Y =$ Young's modulus).

WATCH OUT: use L_0 at the starting temperature; hole dimensions scale like the solid.
CHECK: is the fractional change tiny (of order 10^{-4})?

Question

A steel rail is $12.0\,\mathrm{m}$ long at $5^\circ\mathrm{C}$. Find its length at $45^\circ\mathrm{C}$ ($\alpha = 1.2 \times 10^{-5}\,\mathrm{K}^{-1}$). What stress would develop if the rail could not expand ($Y = 2.0 \times 10^{11}\,\mathrm{Pa}$)?

Predict first

Expansion of the order of millimetres or centimetres?

Set up: $\Delta T = 45 - 5 = 40\,\mathrm{K}$.
Execute: $\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5})(12.0)(40) = 5.8 \times 10^{-3}\,\mathrm{m} = 5.8\,\mathrm{mm}$.
Stress: $Y\alpha\Delta T = (2.0 \times 10^{11})(1.2 \times 10^{-5})(40) = 9.6 \times 10^{7}\,\mathrm{Pa}$.
Evaluate: millimetres, as predicted ($\Delta L/L_0 = 4.8 \times 10^{-4}$, tiny); the stress is large, which is why joints are needed. ✓

Question

A surveyor's steel tape is exactly $50.000\,\mathrm{m}$ long at $20^\circ\mathrm{C}$. (a) What is its length at $35^\circ\mathrm{C}$ ($\alpha = 1.2 \times 10^{-5}\,\mathrm{K}^{-1}$)? (b) At $35^\circ\mathrm{C}$ she reads $35.794\,\mathrm{m}$ from the tape. What is the true distance?

Predict first

Is the change in length of the order of millimetres or centimetres? Is the true distance larger or smaller than the reading?

(a) $\Delta T = 35 - 20 = 15\,\mathrm{K}$;
$\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5})(50.000)(15) = 9.0 \times 10^{-3}\,\mathrm{m} = 9.0\,\mathrm{mm}$;
$L = L_0 + \Delta L = 50.009\,\mathrm{m}$.
(b) Each marked metre is now longer by the factor $L/L_0 = 1 + \alpha\Delta T = 1 + 1.8 \times 10^{-4}$, so the true distance is $(1 + 1.8 \times 10^{-4})(35.794) = 35.800\,\mathrm{m}$.
Evaluate: millimetres, as predicted; the true distance is larger, because the expanded tape reads too small. A $9\,\mathrm{mm}$ difference matters in precision work. ✓

Decide first

1. Equal heat is given to $1\,\mathrm{kg}$ of water and $1\,\mathrm{kg}$ of copper. Which warms more?
2. A larger object expands more than a smaller one of the same material. Why?
3. Can an object absorb heat without its temperature rising?

Answers

1. **The copper.** Its specific heat is about ten times smaller.
2. **More material, more length to expand:** $\Delta L \propto L_0$.
3. **Yes.** During a phase change (next lecture) the energy goes into breaking bonds.

Defining specific heat and heat capacity

Quantity	What it is	What it measures	SI unit	Representation
Specific heat c	the energy needed to raise $1\,\mathrm{kg}$ of a material by $1\,\mathrm{K}$	how much energy a material absorbs per kelvin	$\mathrm{J/(kg\,K)}$	a table value for each material
Heat capacity C	mc, the energy to raise the whole object by $1\,\mathrm{K}$	how much energy a particular object absorbs per kelvin	$\mathrm{J/K}$	a number for the object
Molar heat capacity C_m	the energy to raise one mole by $1\,\mathrm{K}$	the energy per mole per kelvin	$\mathrm{J/(mol\,K)}$	a table value; $C_m = Mc$

Interpret: specific heat answers *how much energy does this material need per kelvin?* Water's large c means it warms slowly and stores much energy.

Rule: heat and specific heat

$$Q = mc\,\Delta T \qquad \text{unit of } c: \ \mathrm{J/(kg\,K)}$$

$Q > 0$: energy transferred *into* the body (its temperature rises). $Q < 0$: energy transferred *out*.

- Specific heat c: the energy per kilogram per kelvin; water: $4186\,\mathrm{J/(kg\,K)}$.
- Heat capacity of an object: $C_{\text{obj}} = mc$.
- Molar heat capacity: with $m = nM$, $Q = nMc\,\Delta T = nC\,\Delta T$, so $C = Mc$ (unit $\mathrm{J/(mol\,K)}$).

Common error

A large c does not make something heat up quickly; it means a *lot* of energy is needed per kelvin. Water heats slowly.

- **TOOLKIT 3 – Heat and specific heat**

1. **IDENTIFY** the body, its mass m and material (c from a table).
2. **PREDICT** the sign of Q: heating $\Rightarrow Q > 0$; cooling $\Rightarrow Q < 0$.
3. $\Delta T = T_f - T_i$ (kelvin or $^\circ$C, the same size).
4. **CALCULATE** $Q = mc\Delta T$ with m in kg and c in $\mathrm{J/(kg\,K)}$.
5. **USE** $Q = C_{\text{obj}}\Delta T$ for an object, or $Q = nC\Delta T$ with the molar heat capacity.

WATCH OUT: this formula holds only if there is no phase change; c is assumed constant over the range.

CHECK: does the sign of Q agree with heating or cooling?

Question

A kettle heats $1.50\,\mathrm{kg}$ of water from $20^\circ\mathrm{C}$ to $100^\circ\mathrm{C}$. (a) How much heat is needed? (b) How long does this take at a heating rate of $2.0\,\mathrm{kW}$ (all energy going into the water)?

Predict first

Order of magnitude of Q: 10^2, 10^5 or $10^8\,\mathrm{J}$? Minutes or hours?

(a) $Q = mc\Delta T = (1.50)(4186)(80) = 5.0 \times 10^5\,\mathrm{J}$. (b)
$t = Q/P = 5.0 \times 10^5/2000 = 2.5 \times 10^2\,\mathrm{s} \approx 4.2\,\mathrm{min}$.
Evaluate: about $10^5\,\mathrm{J}$ and a few minutes, as predicted; $Q > 0$. ✓

Question

During a bout of flu, an $80\,\mathrm{kg}$ man runs a fever of $39.0^\circ\mathrm{C}$ instead of the normal $37.0^\circ\mathrm{C}$. Assuming the human body is mostly water ($c = 4190\,\mathrm{J/(kg\,K)}$), how much heat is needed to raise his temperature by that amount?

Predict first

Order of magnitude of Q: 10^3, 10^5 or $10^7\,\mathrm{J}$? Is it more or less than the kettle in Example 4?

$Q = mc\Delta T = (80)(4190)(2.0) = 6.7 \times 10^5\,\mathrm{J}$.
Evaluate: of order $10^5\,\mathrm{J}$, comparable with the $5.0 \times 10^5\,\mathrm{J}$ for the kettle; $Q > 0$ because the body warms. ✓

Question

An electronic element is made of $23\,\mathrm{mg}$ of silicon. The current through it adds energy at the rate $7.4\,\mathrm{mW} = 7.4 \times 10^{-3}\,\mathrm{J/s}$. If no heat can leave the element, at what rate does its temperature rise? The specific heat of silicon is $705\,\mathrm{J/(kg\,K)}$.

Predict first

Does the temperature rise by a fraction of a kelvin or by tens of kelvin each second?

Set up: divide $Q = mc\Delta T$ by Δt: $\dfrac{\mathrm{d}Q}{\mathrm{d}t} = mc\dfrac{\mathrm{d}T}{\mathrm{d}t}$.

Execute: $\dfrac{\mathrm{d}T}{\mathrm{d}t} = \dfrac{\mathrm{d}Q/\mathrm{d}t}{mc} = \dfrac{7.4 \times 10^{-3}}{(23 \times 10^{-6})(705)} = 0.46\,\mathrm{K/s}$.

Evaluate: a fraction of a kelvin per second, but $0.46\,\mathrm{K/s}$ is $27\,\mathrm{K/min}$, so the element would soon destroy itself. Heat transfer is a design problem. ✓

Common mistakes and the correct habit

Mistake	Instead
Using $Q = mc\Delta T$ across a phase change.	Use $\pm mL$ at the change (Lecture 10).
Believing a large c means quick heating.	It means much energy per kelvin.
Thinking a hole shrinks when the plate is heated.	It expands like the material.
Taking L_0 at the final temperature.	L_0 is the length at the starting temperature.

Contact A: what you can now do

- Distinguish temperature from heat; state the zeroth law and explain a thermometer.
- Convert temperatures and temperature differences; use kelvin when needed.
- Calculate $\Delta L = \alpha L_0 \Delta T$ and $\Delta V = \beta V_0 \Delta T$; find $Q = mc\Delta T$ with the correct sign.

Predict first

Before Wednesday, predict: a hot copper block is dropped into cool water. Is the final temperature closer to the copper's or to the water's? And a metal spoon and a wooden spoon lie in the same room: why does the metal one feel colder?

But what if... two bodies at different temperatures exchange energy, and energy must cross a material? $\rightarrow$ **calorimetry and the three ways heat moves**.

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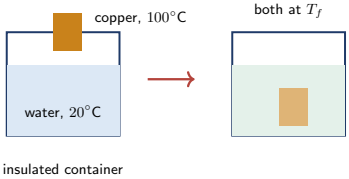
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Contact B: Model, then Apply

How do bodies reach a common temperature, and how does heat move through and between bodies?

Predict first

1. How much heat warms $0.50\,\mathrm{kg}$ of water by $20\,\mathrm{K}$?
   $Q = mc\Delta T = (0.50)(4186)(20) = 4.2 \times 10^4\,\mathrm{J}$.
2. A $2.0\,\mathrm{m}$ steel bar ($\alpha = 1.2 \times 10^{-5}\,\mathrm{K^{-1}}$) warms by $30\,\mathrm{K}$. Find ΔL.
   $\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5})(2.0)(30) = 7.2 \times 10^{-4}\,\mathrm{m} = 0.72\,\mathrm{mm}$.
3. A temperature rises by $15\,\mathrm{K}$. How many degrees Fahrenheit is that? A difference has no offset: $\Delta T_F = \frac{9}{5}(15) = 27^\circ\mathrm{F}$.



Rule: calorimetry

In an isolated system, the heat gained by the cooler parts equals the heat lost by the hotter parts:

$$\sum Q = 0 \quad \text{or} \quad Q_{\text{gained}} = -Q_{\text{lost}}$$

Introduce. A hot body and a cool body sit in an insulated container. They exchange energy until both are at one final temperature T_f.

Explain. Nothing leaves the container, so energy only moves between the bodies. The hot body has $\Delta T < 0$ and $Q < 0$; the cool body has $\Delta T > 0$ and $Q > 0$. This is conservation of energy (Lecture 5) applied to heat.

- **TOOLKIT 4 – Calorimetry (no phase change)**

1. SYSTEM: list every body that exchanges energy; insulate the system from the surroundings.
2. UNKNOWN final temperature T_f (common to all bodies at equilibrium).
3. WRITE $Q_i = m_i c_i (T_f - T_{i,\text{initial}})$ for each body, with signs.
4. SET $\sum Q_i = 0$ and solve for T_f (a linear equation).
5. EVALUATE: T_f must lie between the lowest and highest initial temperatures.

WATCH OUT: if T_f would reach a melting or boiling point, a phase change occurs: use the methods of the next lecture.

CHECK: does the heat lost by the hot body equal the heat gained by the cold one?

Question

A trader pours $0.300\,\text{kg}$ of hot tea, initially at 70.0°C, into a $0.120\,\text{kg}$ aluminium cup initially at 20.0°C. What is the equilibrium temperature? Assume the tea has the same specific heat as water ($4190\,\text{J/(kg K)}$), $c_{\text{Al}} = 910\,\text{J/(kg K)}$, and no heat is exchanged with the surroundings.

Predict first

Will T_f be closer to 70°C or to 20°C? Why? (Compare the heat capacities.)

Execute: $m_t c_w (T_f - 70) + m_{\text{Al}} c_{\text{Al}} (T_f - 20) = 0$

$(1257)(T_f - 70.0) + (109.2)(T_f - 20.0) = 0 \;\Rightarrow\; T_f = \dfrac{87\,990 + 2184}{1257 + 109.2} = 66.0^\circ\text{C}.$

Evaluate: close to 70°C because the tea's heat capacity ($1257\,\text{J/K}$) is much larger than the cup's ($109\,\text{J/K}$). Check: the tea loses $1257(4.0) = 5.0\,\text{kJ}$, the cup gains $109.2(46.0) = 5.0\,\text{kJ}$. ✓

Independent problem

$0.30\,\mathrm{kg}$ of water at $80^\circ\mathrm{C}$ is poured into $0.50\,\mathrm{kg}$ of water at $20^\circ\mathrm{C}$ in an insulated container. Find the final temperature.

Predict first

Will T_f be above or below $50^\circ\mathrm{C}$ (the simple average)? Why?

$$0.30\,c\,(T_f - 80) + 0.50\,c\,(T_f - 20) = 0 \;\Rightarrow\; T_f = \frac{0.30(80) + 0.50(20)}{0.80} = 42.5^\circ\mathrm{C}.$$

Evaluate: below $50^\circ\mathrm{C}$, as predicted, because there is more cold water than hot; between 20 and $80^\circ\mathrm{C}$. ✓

Three ways heat moves

Mechanism	How energy moves	Needs matter?	Example
conduction	collisions and vibrations between neighbouring particles, no bulk motion	yes (a solid, liquid or gas)	a spoon in hot tea
convection	the motion of the fluid itself carries the energy	yes (a fluid)	boiling water, sea breeze
radiation	electromagnetic waves	no	the Sun, a fire

Predict first

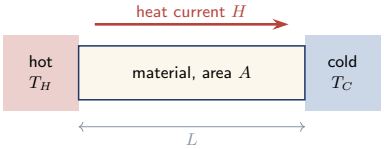
Which mechanism brings the Sun's energy to the Earth? Which warms your hand when you hold it beside a fire, not above it?

Defining the vocabulary of heat transfer

CORE

Quantity	What it is	What it measures	SI unit	Representation
Heat current H	the rate at which energy flows through a material or across a surface	how fast energy is transferred	W	an arrow from hot to cold
Thermal conductivity k	a material property in the conduction law	how well a material conducts energy	$\mathrm{W/(m\,K)}$	a table value
Emissivity e	the ratio of the power radiated to that of an ideal radiator	how well a surface radiates	none	a number from 0 to 1

Interpret: the heat current answers *how fast does the energy move?* Conduction, convection and radiation are three ways to carry it.



Rule: heat current in conduction

$$H = \frac{\mathrm{d}Q}{\mathrm{d}t} = kA\,\frac{T_H - T_C}{L}$$

k: thermal conductivity ($\mathrm{W/(m\,K)}$). H in watts.

- Conduction occurs within a body, or between bodies in contact. It needs a medium and a temperature difference between two points.
- A is the area normal to the flow and L the length along it. Larger A or larger ΔT gives more; a longer path gives less.
- Metals have large k; air, foam and wood have small k (insulators). Insulators *reduce* the rate; they do not stop the flow.

Predict the signs

A copper rod ($k = 385\,\mathrm{W/(m\,K)}$, $A = 2.0 \times 10^{-4}\,\mathrm{m^2}$) lies along x. The end at $x = 0$ is at $100\,^\circ\mathrm{C}$ and the end at $x = 0.50\,\mathrm{m}$ is at $20\,^\circ\mathrm{C}$. What is the sign of $\mathrm{d}T/\mathrm{d}x$, and in which direction does heat flow? What is H?

Predict first

Hint: heat flows from hot to cold. Is the temperature rising or falling along $+x$?

$\mathrm{d}T/\mathrm{d}x = -\dfrac{80}{0.50} = -160\,\mathrm{K/m}$ (negative: T falls with x), so $H = -kA\,\mathrm{d}T/\mathrm{d}x = +(385)(2.0 \times 10^{-4})(160) = +12.3\,\mathrm{W}$ along $+x$, from hot to cold. The minus sign is what makes the heat current point *down* the temperature gradient; it is not a negative conductivity.

But what if the hot end is at $x = 0.50\,\mathrm{m}$? Then $\mathrm{d}T/\mathrm{d}x > 0$ and $H < 0$: the flow is along $-x$. The magnitude form $kA\Delta T/L$ gives the size only, not the direction.

TOOLKIT 5 – Conduction

1. SKETCH the hot side, the cold side, the path length L and the area A perpendicular to the flow.
2. IDENTIFY k and $\Delta T = T_H - T_C$ (in K or $^\circ$C).
3. CALCULATE $H = kA\Delta T/L$ (watts).
4. SERIES: when two layers carry the same current, set $H_1 = H_2$ and solve for the interface temperature.

WATCH OUT: L is the length *along* the flow, and A is the cross-section across it; the current is the same through every layer in steady state.
CHECK: is H positive from hot to cold, and is the interface temperature between T_C and T_H?

Question

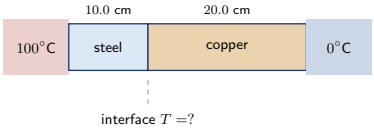
The foam lid of a cooler has area $0.60\,\mathrm{m}^2$ and thickness $2.0\,\mathrm{cm}$ ($k = 0.033\,\mathrm{W/(m\,K)}$). The inside is at $5°\mathrm{C}$ and the outside at $25°\mathrm{C}$. Find the heat current, and the energy conducted in one hour.

Predict first

Is the current of order $1\,\mathrm{W}$, $20\,\mathrm{W}$ or $2000\,\mathrm{W}$?

$H = kA\dfrac{\Delta T}{L} = (0.033)(0.60)\dfrac{20}{0.020} = 20\,\mathrm{W}$. $Q = Ht = (20)(3600) = 7.1\times10^4\,\mathrm{J}$ per hour.

Evaluate: about $20\,\mathrm{W}$, as predicted: small, because k is small. This energy would warm $3.0\,\mathrm{kg}$ of drink by $\Delta T = Q/(mc) = 5.7\,\mathrm{K}$ each hour. ✓



A $10.0\,\mathrm{cm}$ steel bar ($k = 50.2$) is welded to a $20.0\,\mathrm{cm}$ copper bar ($k = 385$); each has a square section $2.00\,\mathrm{cm}$ on a side. The steel's free end is at $100^\circ\mathrm{C}$, the copper's at $0^\circ\mathrm{C}$; the sides are insulated. Find the interface temperature and the heat current.

Predict first

Which bar has the larger temperature drop, copper or steel?

Set up: steady state: the same current in both, $H_{\text{steel}} = H_{\text{Cu}}$, with $A = 4.00 \times 10^{-4}\,\text{m}^2$ for each.

Execute: $\dfrac{50.2\,(100 - T)}{0.100} = \dfrac{385\,(T - 0)}{0.200} \;\Rightarrow\; 50\,200 - 502\,T = 1925\,T \;\Rightarrow\; T = 20.7^{\circ}\text{C}.$

$$H = \frac{kA(T_H - T)}{L} = \frac{(50.2)(4.00 \times 10^{-4})(79.3)}{0.100} = 15.9\,\text{W}.$$

Evaluate: the interface is close to the copper's cold end: most of the temperature drop ($79.3\,\text{K}$) is across the poor conductor (steel), as predicted. Check with copper: $(385)(4.00 \times 10^{-4})(20.7)/0.200 = 15.9\,\text{W}$. ✓

Question

A Styrofoam box for cold drinks has total wall area (including the lid) $0.80\,\mathrm{m}^2$ and wall thickness $2.0\,\mathrm{cm}$ ($k = 0.010\,\mathrm{W/(m\,K)}$). It is filled with ice, water and cans at $0°\mathrm{C}$. (a) What is the rate of heat flow into the box if the outside wall is at $30°\mathrm{C}$? (b) How much ice melts in 3 hours? ($L_f = 3.34 \times 10^5\,\mathrm{J/kg}$; melting is the subject of Lecture 10.)

Predict first

Is H of order 1, 10 or $1000\,\mathrm{W}$?

(a) $H = kA\dfrac{T_H - T_C}{L} = (0.010)(0.80)\dfrac{30-0}{0.020} = 12\,\mathrm{W} = 12\,\mathrm{J/s}.$

(b) $t = 3(3600) = 10\,800\,\mathrm{s}; \quad Q = Ht = (12)(10\,800) = 1.30 \times 10^5\,\mathrm{J};$

$m = \dfrac{Q}{L_f} = \dfrac{1.30 \times 10^5}{3.34 \times 10^5} = 0.39\,\mathrm{kg}.$

Evaluate: about $10\,\mathrm{W}$, as predicted: small, because k is small. Less than half a kilogram of ice melts in 3 hours. ✓

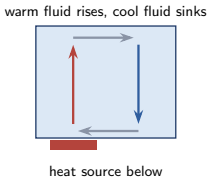
Question

Back to the spoons: both are at $20^\circ\mathrm{C}$, but your hand is at $33^\circ\mathrm{C}$. Explain why the metal one feels colder.

Predict first

Which factor in $H = kA\Delta T/L$ differs between metal and wood?

ΔT is the same for both, but $k_{\text{metal}} \gg k_{\text{wood}}$. The metal carries energy away from your skin much faster, so the skin cools more and the nerves report "colder". The sensation measures the *rate* of energy transfer, not the temperature.

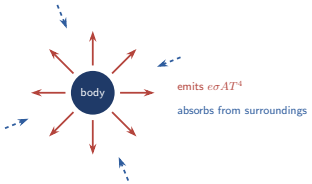


Convection

Energy is carried by the *motion of a fluid*: warm, less dense fluid rises; cooler fluid sinks. It is natural (free) when density differences drive the flow, forced when a blower or pump does. Examples: home heating, the cooling system of an engine, the flow of blood in the body.

There is no simple equation for convection. Three experimental facts:

1. The heat current is directly proportional to the surface area, which is why radiators and cooling fins are large.
2. Viscosity slows natural convection near a stationary surface, leaving a film of air with about the insulating value of $1.3\,\mathrm{cm}$ of plywood. Forced convection thins the film and raises the rate: this is the wind-chill effect.
3. The heat current in free convection is approximately proportional to the $\frac{5}{4}$ power of the temperature difference between the surface and the main body of fluid.



Rule: radiation (Stefan–Boltzmann law)

$$H = e\sigma A\,T^4 \qquad \sigma = 5.67\times10^{-8}\,\mathrm{W/(m^2\,K^4)}$$

e: emissivity ($0 \le e \le 1$); T in **kelvin**. Net exchange with surroundings at T_s: $H_{\text{net}} = e\sigma A\,(T^4 - T_s^4)$.

Define. Radiation is the transfer of heat by electromagnetic waves. It needs no medium: it is how the Sun's energy reaches the Earth. A is the surface area of the radiator and e the emissivity, the ratio of the power radiated to that of an ideal radiator at the same temperature.

Explain. Every body radiates, whatever its temperature; a body at the same temperature as its surroundings still radiates, but absorbs just as much (thermal equilibrium).

- **TOOLKIT 6 – Radiation**

1. IDENTIFY the surface area A, emissivity e and the temperatures of the body and surroundings.
2. CONVERT all temperatures to **kelvin** (Toolkit 1).
3. EMITTED POWER: $H = e\sigma A T^4$.
4. NET POWER with surroundings at T_s: $H_{\text{net}} = e\sigma A(T^4 - T_s^4)$.
WATCH OUT: using $^\circ\text{C}$ in T^4; forgetting the radiation absorbed from the surroundings.
CHECK: is $H_{\text{net}} > 0$ when the body is hotter than its surroundings?

Question

A person has surface area $1.20\,\mathrm{m}^2$ and surface temperature $30^\circ\mathrm{C} = 303\,\mathrm{K}$. The emissivity of the human body is very close to 1 whatever the skin tone. (a) What total power does the body radiate? (b) If the surroundings are at $20^\circ\mathrm{C}$, what is the net rate of radiative heat loss?

Predict first

Order of magnitude: 10, 10^2 or $10^4\,\mathrm{W}$? Is the net loss larger or smaller than (a)?

$T = 303\,\mathrm{K}$, $T_s = 293\,\mathrm{K}$, $e = 1$.
(a) $H = e\sigma AT^4 = (1)(5.67\times10^{-8})(1.20)(303)^4 = 574\,\mathrm{W}$.
(b) $H_{\text{net}} = e\sigma A\,[T^4 - T_s^4] = (5.67\times10^{-8})(1.20)\,[303^4 - 293^4] = 72\,\mathrm{W}$.
Evaluate: the net loss is much smaller, because the person also absorbs from the room, and it is of order $10^2\,\mathrm{W}$, comparable with the body's metabolic power. ✓

Example 12: a steel plate in a forge

PRACTICE

Question

A thin square steel plate, $10\,\text{cm}$ on a side, is heated in a blacksmith's forge to 800°C. If the emissivity is 0.60, what is the total rate at which it radiates energy?

Predict first

Order of magnitude: 10, 10^3 or $10^5\,\text{W}$? Which area do we use: one face or both?

Set up: both faces radiate, so $A = 2(0.10)^2 = 0.020\,\text{m}^2$; $\quad T = 800 + 273 = 1073\,\text{K}$.
Execute: $H = e\sigma AT^4 = (0.60)(5.67 \times 10^{-8})(0.020)(1073)^4 = 9.0 \times 10^2\,\text{W}$.
Evaluate: about $10^3\,\text{W}$, as predicted. The plate is hot, so T^4 is large; a room at 20°C would send back only about $5\,\text{W}$, which is negligible here. ✓

- **TOOLKIT 7 – Rates and energy**

1. RATE: identify the heat current H (W) from conduction (Toolkit 5) or radiation (Toolkit 6).
2. ENERGY: $Q = Ht$ for a constant current over a time t.
3. EFFECT: link to the body: $Q = mc\Delta T$ (Toolkit 3), no phase change.
4. SOLVE for the unknown (t, ΔT or H).

WATCH OUT: H changes if T changes: use the initial rate for small changes only, and say so.
CHECK: are the units consistent: W $\times$ s = J?

Application problem: decide what physics applies

A $0.50\,\mathrm{kg}$ sample of water at $20^\circ\mathrm{C}$ in an insulated container is heated by a copper rod ($k = 385\,\mathrm{W/(m\,K)}$, length $0.40\,\mathrm{m}$, cross-section $2.0\,\mathrm{cm}^2$) whose other end is in a reservoir at $100^\circ\mathrm{C}$. Ignore losses.

(a) What is the initial heat current?
(b) About how long does the water take to warm by $1.0\,\mathrm{K}$?
(c) Is it reasonable to use the initial current for this estimate?

Predict first

IDENTIFY → mechanism and model → PREDICT → EXECUTE → EVALUATE. Minutes or hours?

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4}\,\mathrm{m}^2$.

(a) $H = kA\dfrac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4})\dfrac{80}{0.40} = 15\,\mathrm{W}$.

(b) $Q = mc\Delta T = (0.50)(4186)(1.0) = 2.1 \times 10^3\,\mathrm{J}$;
$t = Q/H = 2093/15.4 = 1.4 \times 10^2\,\mathrm{s} \approx 2.3\,\mathrm{min}$.

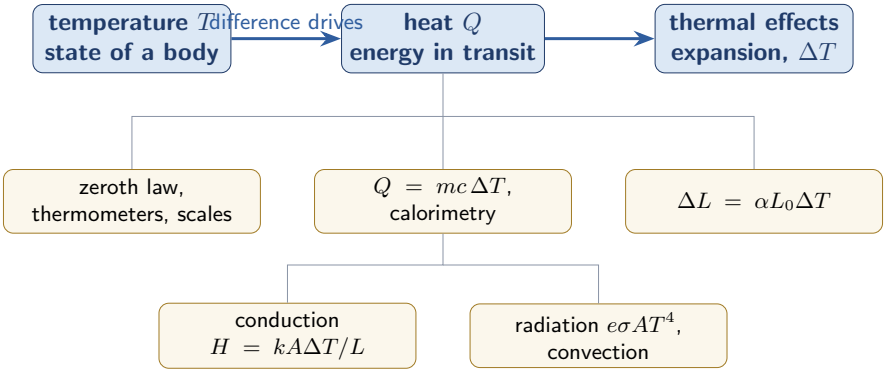
(c) Yes: the water warms by $1\,\mathrm{K}$, so T_C changes from 20 to $21^\circ\mathrm{C}$ and ΔT from 80 to $79\,\mathrm{K}$, a 1% change in H.

Evaluate: minutes, as predicted, for a small temperature change; warming all the way to $100^\circ\mathrm{C}$ would take hours, because the current falls as the water warms. ✓

Common mistakes and the correct habit

Mistake	Instead
Wrong signs in a calorimetry sum.	Write each $Q_i = m_i c_i (T_f - T_i)$ and set $\sum Q = 0$.
Using $^\circ\mathrm{C}$ in T^4.	Kelvin for radiation and ratios.
Forgetting the radiation absorbed from the surroundings.	Use $e\sigma A(T^4 - T_s^4)$.
Using L across the flow in conduction.	L is along the flow; A is across it.
Confusing the heat current H with the energy Q.	$Q = Ht$.
Saying insulators stop the flow.	They reduce the rate; they do not stop it.
Saying the metal spoon is at a lower temperature.	It conducts energy away faster, so it feels colder.

Lecture 9 concept map



MASTER PROCEDURE – THERMAL

1. IDENTIFY The system, the bodies and the process: what is asked?
2. SKETCH The bodies, the boundaries and the direction of energy flow.
3. UNITS Convert temperatures: kelvin for T^4; differences in K or $^\circ$C.
4. PREDICT Sign of Q, direction of heat flow, and the rough size.
5. SELECT THE MODEL Scale, expansion, specific heat, calorimetry, conduction or radiation.
6. EXECUTE Write the equations first, then substitute with units.
7. EVALUATE Signs, units, magnitude, limiting cases, and a second check (energy conservation).

SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: what happens when the temperature reaches a melting or boiling point (phase changes and latent heat).

THE SEVEN TOOLKITS

1	**Temperature scales and conversions**	T against ΔT; kelvin when needed
2	**Thermal expansion**	$\Delta L = \alpha L_0 \Delta T$; $\beta \approx 3\alpha$; thermal stress
3	**Heat and specific heat**	$Q = mc\Delta T$; sign; units
4	**Calorimetry (no phase change)**	$\sum Q = 0$; T_f between the initial values
5	**Conduction**	$H = kA\Delta T/L$; series layers
6	**Radiation**	$H = e\sigma A T^4$; net exchange; kelvin
7	**Rates and energy**	$Q = Ht$ combined with $Q = mc\Delta T$

The Master Procedure sits above these. Earlier toolkits (units, energy and power, signs) are reused, not renamed.

Summary: key ideas

Key ideas

- A body has thermal energy, not heat.
- Use kelvin for T^4 and ratios; differences are equal in K and $^\circ\mathrm{C}$.
- A hole expands with its material.
- Calorimetry is energy conservation: $\sum Q = 0$.
- Conduction rate $\propto A\,\Delta T/L$.
- Radiated power $\propto T^4$.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Scales	$T_F = \frac{9}{5}T_C + 32, \quad T_K = T_C + 273.15$	differences: no offsets
Expansion	$\Delta L = \alpha L_0 \Delta T, \quad \Delta V = \beta V_0 \Delta T$	$\beta \approx 3\alpha$; small ΔT
Heat	$Q = mc\Delta T$	no phase change; J
Calorimetry	$\sum Q = 0$	isolated system
Conduction	$H = kA(T_H - T_C)/L$	steady state; W
Radiation	$H = e\sigma A T^4, \quad H_{\text{net}} = e\sigma A(T^4 - T_s^4)$	kelvin; W
Energy from a rate	$Q = Ht$	constant H

Lectures 10 and 11

Temperature will be linked to the average kinetic energy of molecules; heat Q and work W together change the thermal energy in the first law.

Lecture 12

Heat always flows from hot to cold: this one-way direction is the second law, and sets the limits of engines.

Applications

Insulation, the greenhouse effect, climate and the cooling of the body all use conduction, convection and radiation together.

Physical meaning

Calorimetry is energy conservation (Lecture 5) applied to heat: nothing is lost, only transferred.

- ☐ distinguish temperature from heat?
- ☐ state the zeroth law and explain a thermometer?
- ☐ convert temperatures and temperature differences?
- ☐ calculate linear, area and volume expansion?
- ☐ calculate a thermal stress?
- ☐ find $Q = mc\Delta T$ with the correct sign?
- ☐ solve a calorimetry problem without a phase change?
- ☐ use $H = kA\Delta T/L$, including layers in series?
- ☐ use $H = e\sigma AT^4$ with kelvin?
- ☐ describe convection?
- ☐ combine a rate with a time and $Q = mc\Delta T$?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

The Reference deck holds the supporting slides cut from the live lecture. Tutorial sheet: extra conversion, expansion, specific-heat, calorimetry, conduction, radiation and combined problems. Supplementary notes: the gas thermometer and absolute zero, thermal stress, thermal resistance, and blackbody radiation in outline.

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